

Pivots

- First non-zero entries in each row of the matrix in REF

Example 1

$$\left(\begin{array}{ccccc|c} 1 & 2 & -2 & 0 & 0 & 2 \\ 0 & 1 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 12 & 12 & 24 \end{array} \right) \quad \text{Pivots}$$

Example 2

$$\left(\begin{array}{cc|c} 3 & 5 & 7 \\ 0 & 12 & -9 \\ 0 & 0 & 1 \end{array} \right) \quad \text{Pivots}$$

Contradiction

If pivot is in the final column,
system is inconsistent

Basic variable & Free variables

→ No. of pivots = No. of basic variables

- Unknowns that have a pivot in their column $\Rightarrow$ Basic variables
- Or else $\Rightarrow$ Free variables

$$\left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array} \right) \quad \begin{array}{l} \text{Pivots} \\ \text{Free variables} \\ \text{Basic variables} \end{array}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

Rank

• No. of pivots

↳ No. of basic variables

⇒ Total no. of unknowns = No. of basic variables + No. of free variables

No. of free variables = Total no. of unknowns - Rank A

Note

$\text{Rank}(A) = \text{Rank}(A|b) \Rightarrow$ No Pivot in final column $\Rightarrow$ System consistent

$\text{Rank}(A) \neq \text{Rank}(A|b) \Rightarrow$ Pivot in final column $\Rightarrow$ System inconsistent

$\text{Rank}(A) = \text{No. of unknowns} \Rightarrow$ Unique solution

$\text{Rank}(A) < \text{No. of unknowns} \Rightarrow$ Infinitely many solutions

(
No. of pivots

Overdetermined & Underdetermined

Egn > Unknowns

- No sol.
- Unique sol.
- Infinitely many sol.

Egn < Unknowns

- No sol.
- Infinitely many sol.

Why not unique sol.?

No. of free variables = no. of unknowns - No. of basic variables

= No. of unknowns - No. of pivots

= No. of unknowns - No. of rows

= No. of unknowns - No. of eqn

But underdetermined $\Rightarrow$ Egn < unknown

⇒ No. of free variables ≥ 1 [Since no. of unknowns & eqn can only take integer values]